

Code:

```
#include <stdio.h>
#include <stdlib.h>

int mutex = 1;
int full = 0;
int empty = 10, x = 0;

// Producer function
void producer()
{
    mutex--;
    full++;
    empty--;
    x++;

    printf("\nProducer produces item %d", x);

    mutex++;
}

// Consumer function
void consumer()
{
    mutex--;
    full--;
    empty++;

    printf("\nConsumer consumes item %d", x);
    x--;

    mutex++;
}

int main()
{
    int n;

    while (1)
    {
        printf("\n\n1. Press 1 for Producer");
        printf("\n2. Press 2 for Consumer");
        printf("\n3. Press 3 for Exit");
        printf("\nEnter your choice: ");
        scanf("%d", &n);

        switch (n)
        {
            case 1:
                if ((mutex == 1) && (empty != 0))
                {
                    producer();
                }
                else
                {
                    printf("\nBuffer is full!");
                }
                break;

            case 2:
                if ((mutex == 1) && (full != 0))
                {
                    consumer();
                }
                else
                {
                    printf("\nBuffer is empty!");
                }
                break;

            case 3:
                exit(0);

            default:
                printf("\nInvalid choice!");
        }
    }

    return 0;
}
```

Output:



C:\Users\Dell\Downloads\Unt



```
1. Press 1 for Producer
2. Press 2 for Consumer
3. Press 3 for Exit
Enter your choice: 1
```

```
Producer produces item 1
```

```
1. Press 1 for Producer
2. Press 2 for Consumer
3. Press 3 for Exit
Enter your choice: 2
```

```
Consumer consumes item 1
```

```
1. Press 1 for Producer
2. Press 2 for Consumer
3. Press 3 for Exit
Enter your choice: 3
```

```
Process returned 0 (0x0)    execution time : 15.468 s
Press any key to continue.
```

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